

MASTER OF COMPUTER APPLICATION (MCA)

BMC301: Python Programming

Course Outcome (CO)	Bloom's Knowledge Level (KL)
At the end of course, the student will be able to	
CO 1	Interpret the fundamental Python syntax and semantics and be fluent in the use of Python control flow statements. K1, K2
CO 2	Express proficiency in the handling of strings and functions K1, K2
CO 3	Determine the methods to create and manipulate Python programs by utilizing the data structures like lists, dictionaries, tuples and sets. K3
CO 4	Use OO concepts while programming in Python K1, K2
CO 5	Work with Python using GUI. K4

DETAILED SYLLABUS		
3-0-0		
Unit	Topic	Proposed Lecture
I	Introduction to Python: Introduction and Basics; Setting up path Python Data Variables & Operators: Data Variables and its types, id() and type() functions, Coding Standards, Input-Output: Printing on screen, Reading data from keyboard; Control Structures: if-else, elif, Nested if, Iteration Control structures, Break, Continue & Pass.	08
II	String Manipulation: Accessing Strings, Basic Operations, String slices, Function and Methods. Lists: Introduction, Accessing list, Operations, Working with lists, Function and Methods. Tuple: Introduction, accessing tuples, Operations, Working, Functions and Methods.	08
III	Dictionaries: Introduction, accessing values in dictionaries, Working with dictionaries, Properties, Functions. Functions: Defining & Calling a function, Passing arguments to function – Mutable & Immutable Data Types, Different types of arguments, Recursion, Scope of variables;	08
IV	Modules and Packages: User-defined modules and Standard Library: random, numpy, scipy, sys, Math Module, String Module, List Module, Date & Time Module, Regular Expressions: match, search, replace; File Handling: Introduction, File Types, Creating, Opening, Closing, Renaming, Accessing and deleting files, File pointers, File Modes, Binary files.	08
V	Exception Handling: Exception, Exception Handling, Except clause, Try? finally clause, User Defined Exceptions. Basics of Python for Data Analysis, Introduction to series and dataframes.	08

Suggested Readings:

1. Basin H., "Python for Beginners", New Age International Publishers.
2. Ramalho L., "Fluent Python", SPD.
3. Severance C., "Python for Everybody", SPD.
4. Brown M. C., "The Complete Reference", Mc Graw Hill.
5. Kanetkar Y. and Kanetkar A., "Let Us Python", Bpb.
6. Lutz M., "Learning Python", SPD.